

Here's a comprehensive medical report for a critically ill patient with multiple organ dysfunction syndrome (MODS) following septic shock:

MEDICAL REPORT - CRITICAL CASE

PATIENT INFORMATION

Name: Rajesh Kumar Sharma
Age: 68 years
Gender: Male
Date of Visit: January 9, 2026
Time of Admission: 11:45 PM IST
Medical Record Number: MRN-2026-CR-4782
Admission Status: ICU - Critical Condition

CHIEF COMPLAINT & SYMPTOMS

Primary Presentation: Patient brought to Emergency Department via ambulance in altered mental status with severe respiratory distress, hypotension unresponsive to fluid resuscitation, and multiorgan failure.

Symptom Timeline:

- 72 hours prior: Fever (103.5°F), severe abdominal pain, persistent vomiting, and profuse watery diarrhea (15+ episodes/day)
- 48 hours prior: Increasing confusion, decreased urine output (oliguria), difficulty breathing
- 24 hours prior: Complete disorientation, inability to recognize family members, gasping respirations
- On arrival: Unresponsive to verbal stimuli, cyanotic extremities, agonal breathing pattern

Severity: Life-threatening emergency requiring immediate ICU admission and mechanical ventilation

Associated Symptoms:

- Severe dehydration with sunken eyes and poor skin turgor
- Mottled, cold extremities indicating peripheral shutdown
- Jaundice (yellowing of skin and sclera)
- Petechial rash on trunk and extremities

- Anuria (complete absence of urine output for 18 hours)
 - Disseminated intravascular coagulation (DIC) with bleeding from IV sites
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MEDICAL HISTORY

Past Medical Conditions:

- Chronic Kidney Disease Stage 3 (diagnosed 2019)
- Chronic Obstructive Pulmonary Disease (COPD) - severe, on home oxygen
- Decompensated Cirrhosis (Child-Pugh Class C) secondary to chronic alcoholism
- Type 2 Diabetes Mellitus (poorly controlled, HbA1c 9.8%)
- Coronary Artery Disease with prior STEMI (2022)
- Peripheral Vascular Disease with non-healing diabetic foot ulcer (left foot)
- Chronic Anemia requiring periodic transfusions
- Malnutrition with BMI 16.2 kg/m²

Surgical History:

- Coronary Artery Bypass Graft (CABG) - 2022
- Left below-knee amputation scheduled (cancelled due to current illness)

Current Medications (Home):

- Insulin glargine 20 units subcutaneous daily
- Furosemide 80 mg twice daily
- Spironolactone 50 mg once daily
- Carvedilol 12.5 mg twice daily
- Aspirin 75 mg once daily
- Lactulose 30 mL three times daily
- Rifaximin 550 mg twice daily

Allergies:

- Penicillin (anaphylaxis)
- Sulfa drugs (severe rash)
- Contrast dye (acute kidney injury)

Social History:

- Chronic alcoholism (40+ years, recently abstinent for 6 months)
 - Former smoker (50 pack-years, quit 2020)
 - Lives alone, limited family support
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PHYSICAL EXAMINATION

General Appearance: Critically ill, cachectic male, unresponsive, on mechanical ventilation with multiple life-support systems

Vital Signs (On Admission to ICU)

Parameter	Value	Status
Blood Pressure	68/42 mmHg	Severe Hypotension
Heart Rate	142 bpm	Severe Tachycardia
Temperature	95.2°F (35.1°C)	Severe Hypothermia
Respiratory Rate	38 breaths/min	Severe Tachypnea
SpO ₂	76% on room air	Critical Hypoxemia
SpO ₂ (post-intubation)	88% on FiO ₂ 100%	Persistent Hypoxemia

Glasgow Coma Scale (GCS): 3/15 (E1V1M1) - indicating deep coma

Cardiovascular: Thready, barely palpable peripheral pulses; cold, clammy, mottled skin; delayed capillary refill >5 seconds; loud S3 gallop; jugular venous distension

Respiratory: Bilateral coarse crackles throughout lung fields, decreased air entry bilaterally, using accessory muscles, intercostal retractions

Abdominal: Grossly distended with tense ascites, absent bowel sounds, diffuse tenderness with guarding and rigidity, liver edge palpable 6 cm below costal margin

Neurological: Unresponsive to painful stimuli, absent deep tendon reflexes, fixed dilated pupils (6mm bilateral), no spontaneous movements

Extremities: Severe edema (4+ pitting) bilateral lower limbs, gangrenous changes on left foot with purulent discharge, multiple pressure ulcers (Stage IV sacral ulcer)

Skin: Icteric (deeply jaundiced), widespread petechiae and ecchymoses, cool and mottled appearance

LABORATORY RESULTS

Complete Blood Count (CBC)

Test	Result	Normal Range	Interpretation
Hemoglobin	6.2 g/dL	13.0-17.0 g/dL	Severe Anemia
Hematocrit	18.5%	40-52%	Critically Low
WBC Count	2,100 cells/ μ L	4,000-11,000 cells/ μ L	Severe Leukopenia
Neutrophils	68%	40-70%	Normal
Lymphocytes	12%	20-40%	Lymphopenia
Platelets	28,000/ μ L	150,000-400,000/ μ L	Severe Thrombocytopenia

Comprehensive Metabolic Panel

Test	Result	Normal Range	Interpretation
Sodium	162 mEq/L	135-145 mEq/L	Severe Hyponatremia
Potassium	6.8 mEq/L	3.5-5.0 mEq/L	Life-threatening Hyperkalemia
Chloride	112 mEq/L	98-106 mEq/L	Hyperchloremia
Bicarbonate	12 mEq/L	22-28 mEq/L	Severe Metabolic Acidosis
BUN	142 mg/dL	7-20 mg/dL	Severe Azotemia
Creatinine	8.4 mg/dL	0.6-1.2 mg/dL	Acute Kidney Injury Stage 3
eGFR	6 mL/min/1.73m ²	>60 mL/min/1.73m ²	End-Stage Renal Disease
Glucose	418 mg/dL	70-100 mg/dL	Severe Hyperglycemia
Calcium	7.2 mg/dL	8.5-10.5 mg/dL	Hypocalcemia

Liver Function Tests

Test	Result	Normal Range	Interpretation
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Total Bilirubin	18.6 mg/dL	0.3-1.2 mg/dL	Severe Hyperbilirubinemia
Direct Bilirubin	12.4 mg/dL	0.0-0.3 mg/dL	Conjugated Hyperbilirubinemia
AST (SGOT)	1,842 U/L	10-40 U/L	Massive Hepatocellular Injury
ALT (SGPT)	1,564 U/L	7-56 U/L	Massive Hepatocellular Injury
Alkaline Phosphatase	486 U/L	44-147 U/L	Severely Elevated
Total Protein	4.2 g/dL	6.0-8.0 g/dL	Severe Hypoproteinemia
Albumin	1.8 g/dL	3.5-5.0 g/dL	Severe Hypoalbuminemia
INR	4.8	0.8-1.2	Critical Coagulopathy
Ammonia	248 µg/dL	15-45 µg/dL	Severe Hyperammonemia

Cardiac Biomarkers

Test	Result	Normal Range	Interpretation
Troponin I	8.42 ng/mL	<0.04 ng/mL	Acute Myocardial Injury

CK-MB	186 U/L	<25 U/L	Cardiac Muscle Damage
NT-proBNP	18,400 pg/mL	<125 pg/mL	Severe Heart Failure

Coagulation Profile

Test	Result	Normal Range	Interpretation
PT	42.6 seconds	11-13.5 seconds	Severely Prolonged
APTT	88.4 seconds	25-35 seconds	Severely Prolonged
D-Dimer	24,800 ng/mL	<500 ng/mL	DIC/Massive Thrombosis
Fibrinogen	82 mg/dL	200-400 mg/dL	Severe Hypofibrinogenemia

Arterial Blood Gas (ABG)

Parameter	Result	Normal Range	Interpretation
pH	6.98	7.35-7.45	Severe Acidemia
PaCO ₂	68 mmHg	35-45 mmHg	Severe Hypercapnia

PaO ₂	52 mmHg	80-100 mmHg	Severe Hypoxemia
HCO ₃ ⁻	12 mEq/L	22-26 mEq/L	Metabolic Acidosis
Lactate	14.8 mmol/L	0.5-2.2 mmol/L	Severe Lactic Acidosis

Inflammatory Markers & Infection Studies

Test	Result	Normal Range	Interpretation
Procalcitonin	48.6 ng/mL	<0.5 ng/mL	Severe Sepsis/Septic Shock
CRP	386 mg/L	<10 mg/L	Severe Inflammation
ESR	98 mm/hr	0-15 mm/hr	Severely Elevated

Blood Culture Results

- Preliminary (24 hours): Gram-negative rods growing in 3/3 aerobic bottles
- Final (48 hours): *Escherichia coli* (Extended Spectrum Beta-Lactamase producing) - resistant to multiple antibiotics
- Sensitivity: Sensitive only to Meropenem and Colistin

IMAGING STUDIES

Chest X-Ray (Portable AP View)

Date: January 9, 2026, 11:58 PM

Findings:

- Bilateral diffuse alveolar infiltrates consistent with Acute Respiratory Distress Syndrome (ARDS)
- Moderate bilateral pleural effusions
- Cardiomegaly (cardiothoracic ratio 0.68)
- Endotracheal tube in proper position (3 cm above carina)
- Central venous catheter tip in superior vena cava
- No pneumothorax identified

Impression: Severe ARDS with pulmonary edema and bilateral pleural effusions

CT Abdomen and Pelvis (Non-Contrast Emergency Protocol)

Date: January 10, 2026, 12:45 AM

Findings:

- Massive ascites with fluid density measuring 18-22 HU
- Hepatomegaly with cirrhotic liver (nodular contour, irregular margins)
- Portal vein thrombosis
- Splenomegaly (18 cm in longest dimension)
- Bilateral hydronephrosis with cortical thinning
- Thickened, edematous bowel loops suggestive of enterocolitis
- Free air under diaphragm indicating bowel perforation
- Extensive intra-abdominal fluid collections with air-fluid levels (abscesses)

Impression: Perforated viscus with peritonitis, cirrhotic liver with portal hypertension, bilateral obstructive uropathy

Echocardiography (Bedside Transthoracic)

Date: January 10, 2026, 1:15 AM

Findings:

- Severe global left ventricular systolic dysfunction (LVEF 18-20%)
- Moderate to severe mitral regurgitation
- Moderate tricuspid regurgitation with estimated RVSP 58 mmHg
- Dilated inferior vena cava with minimal respiratory variation
- Small to moderate pericardial effusion (no tamponade)
- Regional wall motion abnormalities in LAD territory

Impression: Severe cardiomyopathy with multivalvular dysfunction and evidence of right heart strain

CT Head (Non-Contrast)

Date: January 10, 2026, 1:30 AM

Findings:

- Diffuse cerebral edema with effacement of sulci
- Loss of grey-white matter differentiation
- Compression of lateral ventricles
- No acute hemorrhage or mass lesion
- Multiple lacunar infarcts in basal ganglia (chronic)

Impression: Severe cerebral edema consistent with hepatic encephalopathy Grade IV and hypoxic-ischemic injury

CLINICAL NOTES & ASSESSMENT

Critical Care Assessment by Dr. Priya Menon, MD (Intensivist)

Date/Time: January 10, 2026, 2:00 AM IST

Primary Diagnosis: Septic Shock with Multiple Organ Dysfunction Syndrome (MODS)

Contributing Diagnoses:

1. Severe sepsis from intra-abdominal source (*E. coli* bacteremia, ESBL+)
2. Perforated bowel with diffuse peritonitis and intra-abdominal abscesses
3. Acute Respiratory Distress Syndrome (ARDS) - severe (PaO₂/FiO₂ ratio: 52)
4. Acute Kidney Injury Stage 3 (AKIN criteria) on chronic kidney disease
5. Acute-on-chronic liver failure (decompensated cirrhosis)
6. Disseminated Intravascular Coagulation (DIC)
7. Hepatic encephalopathy Grade IV (coma)
8. Cardiogenic shock with acute heart failure exacerbation
9. Severe metabolic acidosis with lactic acidosis
10. Acute myocardial injury (likely sepsis-induced cardiomyopathy)

Clinical Observations

This is an extremely critical case with grave prognosis. The patient presents with catastrophic multi-system failure involving simultaneous dysfunction of respiratory, cardiovascular, renal, hepatic, hematological, and neurological systems.

Septic Shock Features:

- Refractory hypotension despite aggressive fluid resuscitation (4 liters crystalloid administered)
- Requiring high-dose vasopressor support (Norepinephrine 0.8 mcg/kg/min + Vasopressin 0.04 units/min)
- Serum lactate >14 mmol/L indicating severe tissue hypoperfusion
- Procalcitonin levels extremely elevated confirming bacterial sepsis

Respiratory Failure:

- Severe ARDS requiring mechanical ventilation with lung-protective strategies
- FiO₂ requirement 100% with persistently poor oxygenation
- Bilateral infiltrates on imaging with P/F ratio <100 (severe ARDS criteria)

Renal Failure:

- Complete anuria for 18+ hours
- Creatinine elevated to 8.4 mg/dL from baseline 2.1 mg/dL
- Severe electrolyte derangements (life-threatening hyperkalemia at 6.8 mEq/L)
- Urgent continuous renal replacement therapy (CRRT) initiated

Hepatic Failure:

- Fulminant hepatic dysfunction with coagulopathy (INR 4.8)
- Grade IV hepatic encephalopathy (Glasgow Coma Scale 3)
- Massive elevation of transaminases (AST >1800, ALT >1500)
- Severe jaundice (bilirubin 18.6 mg/dL)
- Ammonia levels critically elevated

Hematological Crisis:

- Disseminated intravascular coagulation with consumption coagulopathy
- Severe thrombocytopenia (28,000/μL) with active bleeding
- Profound anemia requiring urgent transfusion
- Leukopenia indicating bone marrow suppression/overwhelming sepsis

Cardiovascular Collapse:

- Cardiogenic component to shock (LVEF 18-20%)
- Elevated cardiac biomarkers suggesting acute myocardial injury
- Severe heart failure with pulmonary edema

Surgical Emergency:

- Bowel perforation with free intraperitoneal air
- Diffuse peritonitis with multiple abscesses
- Patient deemed too unstable for emergency laparotomy - prohibitive surgical risk

Sequential Organ Failure Assessment (SOFA) Score: 18/24

- Indicates >95% predicted mortality

APACHE II Score: 38

- Predicted mortality >85%

Prognosis

EXTREMELY POOR - The combination of septic shock, MODS involving six organ systems, advanced underlying chronic diseases, and inoperability of surgical pathology confer a mortality risk exceeding 95%. Despite maximal intensive care support, survival is highly unlikely.

Current Management (Intensive Care Protocol)

Respiratory Support:

- Mechanical ventilation: Volume control mode
- Tidal volume: 6 mL/kg ideal body weight (ARDS protocol)
- PEEP: 14 cm H₂O
- FiO₂: 100%
- Prone positioning initiated

Hemodynamic Support:

- Norepinephrine infusion 0.8 mcg/kg/min
- Vasopressin 0.04 units/min
- Central venous catheter for monitoring (CVP 18 mmHg)
- Arterial line for continuous BP monitoring
- Goal MAP >65 mmHg (currently not achieved)

Antibiotic Therapy (Adjusted for Culture):

- Meropenem 2g IV every 8 hours (ESBL coverage)
- Colistin 300 mg IV loading, then 150 mg IV q12h
- Metronidazole 500 mg IV q8h (anaerobic coverage)
- Fluconazole 400 mg IV daily (fungal prophylaxis)

Renal Replacement:

- Continuous Renal Replacement Therapy (CRRT) initiated
- Bicarbonate-buffered dialysate for acid-base correction

Blood Product Transfusion:

- 4 units Packed RBCs transfusing

- 6 units Fresh Frozen Plasma for coagulopathy
- 2 units Platelet concentrate
- Cryoprecipitate for fibrinogen replacement

Metabolic Support:

- Insulin infusion for hyperglycemia control
- Calcium gluconate for hypocalcemia
- Sodium bicarbonate for severe acidosis
- Stress ulcer prophylaxis (pantoprazole IV)

Monitoring:

- Continuous cardiac monitoring
- Hourly vital signs
- Arterial blood gas every 2 hours
- Strict intake-output monitoring
- Neurological assessment

Family Discussion

Extensive discussion held with patient's son regarding extremely grave prognosis. Family informed of multi-organ failure, life-threatening condition, and very low probability of survival despite maximal therapy. Do Not Resuscitate (DNR) order discussed - family requested full resuscitative measures continue for 48 hours, then reassess. Palliative care consultation requested.

Consultant Recommendations

- Surgery: Patient not a surgical candidate due to hemodynamic instability and prohibitive operative risk
- Nephrology: Continue CRRT, poor renal recovery expected
- Cardiology: Medical management only, no interventional options
- Gastroenterology/Hepatology: End-stage liver disease, not a transplant candidate
- Palliative Care: Consulted for goals of care discussion

Attending Physician: Dr. Priya Menon, MD, FCCP

Intensivist & Critical Care Specialist

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Contact: +91-484-2567890

Report Generated: January 10, 2026, 2:31 AM IST

Hospital: Apollo Multispecialty Hospital, Kochi, Kerala

CRITICAL ALERT: This patient requires continuous intensive monitoring and is at EXTREME risk of imminent death.